

Xinyang Han

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Education

University of California, Berkeley | EECS Department

Berkeley, CA, USA

Concurrent Student in EECS

Jan. 2023 – Oct. 2023

GPA: **4.00** / 4.00

University of Chinese Academy of Science (UCAS) | CS Department

Beijing, PRC

B.Eng. in Computer Science and Technology

Sep. 2020 – Present.

GPA: **3.83** / 4.00

Course Related to My Research Interests

- **Computer Sciences:** 3D Vision Seminar(A), Artificial Intelligence(A), Data Structure(A-), Introduction to Theory of Computation, Computer Graphics(A), Introduction to Algorithm and Theory of Computing, Computer Network Security(A).
- **Mathematics:** Discrete Mathematics(A), Probability and Mathematical Statistics(A), Linear Algebra(A-), Calculus(A)

Publication

[1] **Xinyang Han***, Zelin Gao*, Angjoo Kanazawa, Shubham Goel, Yossi Gandelsman. *SAP3D: The More You See in 2D, the More You Perceive in 3D.*

Submitted to the Conference on Computer Vision and Pattern Recognition. (CVPR2024)

[2] Xudong Wang*, Dantong Niu*, **Xinyang Han***, Long Lian, Roei Herzig, Trevor Darrell. *Unsupervised Universal Image Segmentation.*

Submitted to the Conference on Computer Vision and Pattern Recognition. (CVPR2024)

[3] Yujie Yuan*, **Xinyang Han***, Yue He, Fanglue Zhang, Lin Gao. *MuNeRF: Robust Makeup Transfer in Neural Radiance Fields.*

Submitted to the IEEE Transactions on Visualization and Computer Graphics. (TVCG)

[4] Shu-Yu Chen, Yue-Ren Jiang, Hongbo Fu, **Xinyang Han**, Zitao Liu, Rong Li, Lin Gao. *DeepFaceReshaping: Interactive Deep Face Reshaping via Landmark Manipulation.*

International Conference on Computational Visual Media. (CVM2023)

[5] Tianyu Zhang, Weiqing Min, **Xinyang Han**, Shuqiang Jiang, Yong Rui. *A Review on Action Prediction in Videos.*

Chinese Journal of Computers. (CJC2023)

Research Experience

[1] Unsupervised Pretraining on Gigapixel WSIs For Cancer Prediction.

UC Berkeley

Advisor: Prof. Adam Yala

Nov. 2023 – Dec. 2023

- Established an efficient framework for unsupervised pretraining on gigapixel WSIs for cancer prediction.

[2] SAP3D: The More You See in 2D, the More You Perceive in 3D.

UC Berkeley

Advisor: Prof. Angjoo Kanazawa

Jan. 2023 – Oct. 2023

- The first work to reconstruct high-quality 3D objects from very few unposed images (e.g., 3 images).
- Scaled a pose estimation model and used the prior in pose-guided generative model to enable sparse pose estimation.

[3] Unsupervised Universal Image Segmentation.

UC Berkeley

Advisor: Prof. Trevor Darrell

July. 2023 – Sep. 2023

- Established a unified framework, U2Seg, for unsupervised image segmentation, including instance, semantic, and panoptic segmentation, improving performance over all specialized methods in each category.
- Set a new benchmark for unsupervised panoptic segmentation.

[4] Robust Makeup Transfer in Neural Radiance Field.

Institute of Computing Technology, CAS

Advisor: Prof. Lin Gao

Feb. 2022 – Dec. 2022

- Built up a 3D consistent makeup transfer pipeline using dynamic neural radiance field for head representation that outperform prev. SOTA methods.

- Created a comprehensive makeup supervisor for robust, detailed makeup transfer.

[5] Interactive Deep Face Reshaping via Landmark Manipulation. Institute of Computing Technology, CAS

Advisor: Prof. Lin Gao

March. 2022 – July. 2022

- Developed DeepFaceReshaping, integrating landmark-driven techniques for face reshaping in deep learning models.
- Enhanced facial component manipulation using neural shape deformation.
- Introduced a Transformer-based network for effective image synthesis after landmark editing.
- Improved face reshaping accuracy and user experience, surpassing prev. SOTA methods.

[6] A Review on Action Prediction in Videos.

Institute of Computing Technology, CAS

Advisor: Prof. Shuqiang Jiang

Nov. 2021 – Feb. 2022

- Summarized the framework for video action prediction, differentiating short-term and long-term modalities.
- Categorized methods by structure, modality, and strategy.
- Benchmarked prevalent datasets and analyzed method performance.
- Highlighted future research avenues, emphasizing dataset enhancement and optimized learning.

Awards and Honors

UCAS - Outstanding Student Leader (top 5%)	Jun 2021
UCAS - Excellent Student Award (top 5%)	Jun 2021
Beijing Challenge Cup - Grand Prize (top 1%)	May 2021
Internet Innovation and Entrepreneurship Competition - Second Prize (top 5%)	Aug 2021
UCAS - Excellent Cadre Recognition	Aug 2021
Science and Art Fusion Creation Contest - Third Prize	Dec 2021
UCAS Scientific Fiction Contest - Excellent Award	Dec 2021

Extracurricular Activities

UCAS Student Union - Deputy Director, Organization Department	Dec 2021 - Aug 2022
Innovation and Entrepreneurship Team - Leader	Nov 2020 - Sep 2022
UCAS - Class Monitor	Dec 2020 - Aug 2021

Skills

Programming Languages: Python, PyTorch, TensorFlow, C++, Jax, Javascript, HTML

Languages: Chinese Mandarin (Native), English (TOEFL Score: 106)